Name Surname: KEY 14.12.2022

ID:

Signature:

CSE4065 Fall 2022 Quiz II (Duration: 40 min)

Q1) (30 pts.) Use ClustalW to apply multiple sequence alignment to the following 3 sequences. Show all your steps including guide tree creation and alignment order.

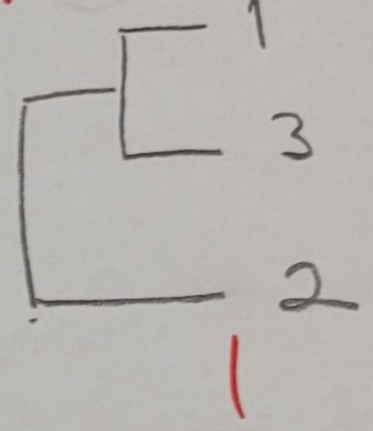
S1: AACG
S2: ACTC
S3: AACTG

S1,2 = A A C G –
A – C T C
✓ ✓

⇒ 2/5 – 5 pts.

S1,3 = A A C – G
A A C T G

⇒ 4/5 – 5 pts.



S1,3 = A A C -/t G
A – C T C

S2,3 = A – C T C
A A C T G
✓ ✓✓

⇒ 3/5 – 5 pts.

5 pts.

S1,2,3 =

A	A	C	–	G
A	A	C	T	G
A	–	C	T	C

– 10 pts.

Q2) (20 pts.) Consider the following alignment:

ATTGTTGCT

ATC---GCT
✓✓ ✓✓✓

a) What is the score of this global alignment if the match score is 2, the mismatch penalty is 1, and the indel penalty is affine with a gap opening penalty of 3 and a gap extension penalty of 1?

Score = 5 * 2 – 1 – 3 – 2 = 4 → 10 pts.
10 – 6

b) Apply local alignment, what is the score? Also show the alignment.

A	T	T	G	T	T
A	–	–	–	T	C

G	C	T	
G	C	T	

→ Score = 7 – 10 pts.

Score = 6 – 8 pts.

1: 0 1 | 7 | 5 6 | 4 3 | 8 | 2 | 9 $b(\pi)=6$

2: 0 1 | 7 | 5 6 | 4 3 2 | 8 9 $b(\pi)=4$

3: 0 1 2 3 4 | 6 5 | 7 8 9 $b(\pi)=2$

4: 0 1 2 3 4 5 6 7 8 9 $b(\pi)=0$

Q3) (30 pts.) Compute the number of steps required by the given algorithm to sort the following permutation (i.e., to transform it into the identity permutation.) Show your work in each step.

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a) Greedy Algorithm – Simple Reversal Sort

step 0: 4 1 2 5 3

1: 1 4 2 5 3

2: 1 2 4 5 3

3: 1 2 3 5 4

4: 1 2 3 4 5

4 steps

(15)

b) Improved Break Point Reversal Sort

step 0: 0 | 4 | 1 2 | 5 | 3 | 6 $b(\pi)=5$

1: 0 | 4 | 1 2 3 | 5 6 $b(\pi)=3$

2: 0 | 4 3 2 1 | 5 6 $b(\pi)=2$

3: 0 1 2 3 4 5 6 $b(\pi)=0$

(15)

3 steps

4 steps: 10

Q4) (20 pts.) Perform the breakpoint reversal sort algorithm with P = 3 4 6 5 7 1 8 2

as the input and show all intermediate permutations. Is this the optimal solution? Why?

step 0: 0 | 3 4 | 6 5 | 7 | 1 | 8 | 2 | 9 $b(\pi)=7$

1: 0 | 3 4 5 6 7 | 1 | 8 | 2 | 9 $b(\pi)=5$

2: 0 | 3 4 5 6 7 8 | 1 2 | 9 $b(\pi)=3$

3: 0 | 2 1 | 8 7 6 5 4 3 | 9 $b(\pi)=3$

4: 0 1 2 | 8 . . . 3 | 9 $b(\pi)=2$

5: $b(\pi)=0$